1) a)

$x = 4e^{-t} + 2e^{-2t}$

The solution corresponds to the auxiliary equation having roots -1 and -2, which are distinct real negative roots.

Therefore, the differential equation is modelling Heavy Damping (Overdamping)

1) b)

$x = e^{-3t}(4\cos t - 3\sin t)$

The solution corresponds to the auxiliary equation having roots $-3 \pm i$, which are complex roots with negative real part.

Therefore, the differential equation is modelling Light Damping (Underdamping)

1) c)

$x = 7te^{-4t} + 4e^{-4t}$

The solution corresponds to the auxiliary equation having a repeated real negative root, -4.

Therefore, the differential equation is modelling Critical Damping

1) d)

$x = 2e^{-6t} + 5e^{3t}$

The solution corresponds to the auxiliary equation having roots -6 and 3. Since one root is positive, the solution contains a growing exponential term.

Therefore, the differential equation models neither Heavy Damping (Overdamping), Critical Damping, or Light Damping (Underdamping).

1) e)

$x = e^{4t}(5\cos 2t + 8\sin 2t)$

The solution corresponds to the auxiliary equation having roots $4 \pm 2i$, which are complex roots with positive real part. As the real part is positive, the solution contains a growing exponential term.

Therefore, the differential equation models neither Heavy Damping (Overdamping), Critical Damping, or Light Damping (Underdamping).

1) f)

$x = 9\mathrm{e}^{-t} - 3t\mathrm{e}^{-t}$

The solution corresponds to the auxiliary equation having a repeated real negative root, -1.

Therefore, the differential equation is modelling Critical Damping

1) g)

$x = \mathrm{e}^{-7t}\left(3\sin\frac{t}{2} - 4\cos\frac{t}{2}\right)$

The solution corresponds to the auxiliary equation having roots $-7 \pm \frac{1}{2}\mathrm{i}$, which are complex roots with negative real part.

Therefore, the differential equation is modelling Light Damping (Underdamping)

1) h)

$x = 7\mathrm{e}^{-6t} - 10\mathrm{e}^{-12t}$

The solution corresponds to the auxiliary equation having roots -6 and -12, which are distinct real negative roots.

Therefore, the differential equation is modelling Heavy Damping (Overdamping)

1) i)

$x = 8t\mathrm{e}^{t} - 3\mathrm{e}^{t}$

The solution corresponds to the auxiliary equation having a repeated real positive root, 1. Since the root is positive, the solution contains two growing exponential terms.

Therefore, the differential equation models neither Heavy Damping (Overdamping), Critical Damping, or Light Damping (Underdamping).

1) j)

$x = 5\mathrm{e}^{-4t}\cos 6t$

The solution corresponds to the auxiliary equation having roots $-4 \pm 6\mathrm{i}$, which are complex roots with negative real part.

Therefore, the differential equation is modelling Light Damping (Underdamping)

2) a)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 6\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 0$

Auxiliary Equation:

$m^2 + 6m + 5 = 0$

$\Rightarrow (m+1)(m+5) = 0$

$\Rightarrow m = -1$ or $m = -5$

General Solution:

$y = A\mathrm{e}^{-x} + B\mathrm{e}^{-5x}$

This is Heavy Damping (Overdamping)

2) b)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$

Auxiliary Equation:

$m^2 + 4m + 4 = 0$

$\Rightarrow (m+2)^2 = 0$

$\Rightarrow m = -2$

General Solution:

$y = Ax\mathrm{e}^{-2x} + B\mathrm{e}^{-2x} = (Ax + B)\mathrm{e}^{-2x}$

This is Critical Damping

2) c)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 0$

Auxiliary Equation:

$m^2 + 2m + 5 = 0$

$\Rightarrow (m+1)^2 - 1 + 5 = 0$

$\Rightarrow (m+1)^2 = -4$

$\Rightarrow m = -1 \pm 2\mathrm{i}$

General Solution:

$y = \mathrm{e}^{-x}(A\cos 2x + B\sin 2x)$

This is Light Damping (Underdamping)

2) d)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + y = 0$

Auxiliary Equation:

$m^2 + 2m + 1 = 0$

$\Rightarrow (m+1)^2 = 0$

$\Rightarrow m = -1$

General Solution:

$y = Ax\mathrm{e}^{-x} + B\mathrm{e}^{-x} = (Ax + B)\mathrm{e}^{-x}$

This is Critical Damping

2) e)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 11\frac{\mathrm{d}y}{\mathrm{d}x} + 30y = 0$

Auxiliary Equation:

$m^2 + 11m + 30 = 0$

$\Rightarrow (m+5)(m+6) = 0$

$\Rightarrow m = -5$ or $m = -6$

General Solution:

$y = A\mathrm{e}^{-5x} + B\mathrm{e}^{-6x}$

This is Heavy Damping (Overdamping)

2) f)

$2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 7\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 0$

Auxiliary Equation:

$2m^2 + 7m + 5 = 0$

$\Rightarrow (2m+5)(m+1) = 0$

$\Rightarrow m = -\frac{5}{2}$ or $m = -1$

General Solution:

$y = A\mathrm{e}^{-\frac{5}{2}x} + B\mathrm{e}^{-x}$

This is Heavy Damping (Overdamping)

2) g)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 13y = 0$

Auxiliary Equation:

$m^2 + 4m + 13 = 0$

$\Rightarrow (m+2)^2 - 4 + 13 = 0$

$\Rightarrow (m+2)^2 = -9$

$\Rightarrow m = -2 \pm 3\mathrm{i}$

General Solution:

$y = \mathrm{e}^{-2x}(A\cos 3x + B\sin 3x)$

This is Light Damping (Underdamping)

2) h)

$2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 6\frac{\mathrm{d}y}{\mathrm{d}x} + 17y = 0$

Auxiliary Equation:

$2m^2 + 6m + 17 = 0$

$\Rightarrow 2(m^2 + 3m) + 17 = 0$

$\Rightarrow 2\left(\left(m + \frac{3}{2}\right)^2 - \frac{9}{4}\right) + 17 = 0$

$\Rightarrow 2\left(m + \frac{3}{2}\right)^2 - \frac{9}{2} + 17 = 0$

$\Rightarrow \left(m + \frac{3}{2}\right)^2 = -\frac{25}{4}$

$\Rightarrow m = -\frac{3}{2} \pm \frac{5}{2}\mathrm{i}$

General Solution:

$y = \mathrm{e}^{-\frac{3}{2}x}\left(A\cos\frac{5}{2}x + B\sin\frac{5}{2}x\right)$

This is Light Damping (Underdamping)

2) i)

$9\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 30\frac{\mathrm{d}y}{\mathrm{d}x} + 25y = 0$

Auxiliary Equation:

$9m^2 + 30m + 25 = 0$

$\Rightarrow (3m + 5)^2 = 0$

$\Rightarrow m = -\frac{5}{3}$

General Solution:

$y = Ax\mathrm{e}^{-\frac{5}{3}x} + B\mathrm{e}^{-\frac{5}{3}x} = (Ax + B)\mathrm{e}^{-\frac{5}{3}x}$

This is Critical Damping

3) a)

This is Heavy Damping (Overdamping) where $k^2 > 4\omega^2$

The auxiliary equation has two distinct real roots, both of which are negative.

3) b)

This is Light Damping (Underdamping) where $k^2 < 4\omega^2$

The auxiliary equation has two complex conjugate roots, where the real part is negative.

4) a)

$y = 3e^{-x} - 2e^{-2x}$

$\frac{dy}{dx} = -3e^{-x} + 4e^{-2x}$

Stationary points exist when $\frac{dy}{dx} = 0$

$-3e^{-x} + 4e^{-2x} = 0$

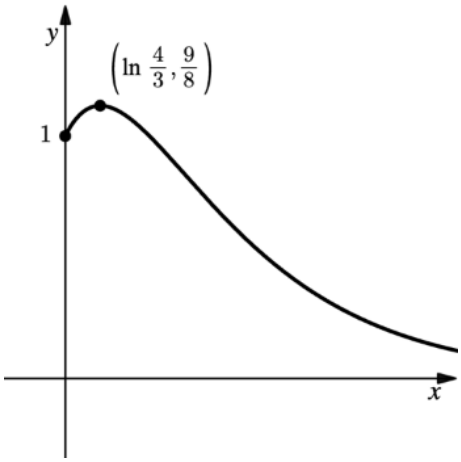
$\Rightarrow 4e^{-2x} = 3e^{-x}$

$\Rightarrow \frac{4}{3} = e^{x}$

$\Rightarrow x = \ln\frac{4}{3}$

$y|_{x=\ln\frac{4}{3}} = \frac{9}{8}$

So there is a stationary point at $\left(\ln\frac{4}{3}, \frac{9}{8}\right)$



4) b)

$y = 4e^{-3x} + 3e^{-x}$

$\frac{dy}{dx} = -12e^{-3x} - 3e^{-x}$

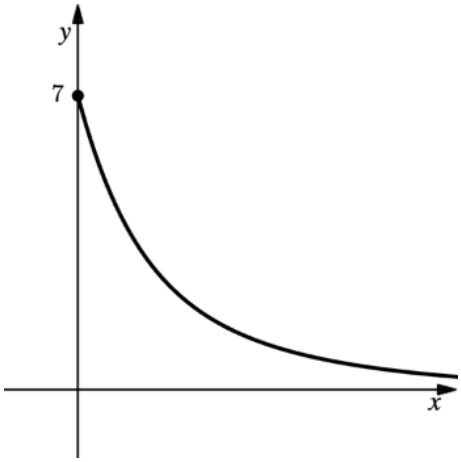
Stationary points exist when $\frac{dy}{dx} = 0$

$-12e^{-3x} - 3e^{-x} = 0$

$\Rightarrow -4e^{-3x} = e^{-x}$

$\Rightarrow -4 = e^{2x}$

This has no real solutions, so there are no stationary points.



4) c)

$y = -3e^{-2x} - 6e^{-4x}$

$\frac{dy}{dx} = 6e^{-2x} + 24e^{-4x}$

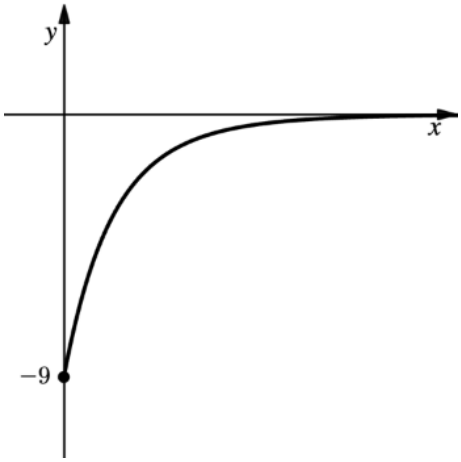
Stationary points exist when $\frac{dy}{dx} = 0$

$6e^{-2x} + 24e^{-4x} = 0$

$\Rightarrow e^{-2x} = -4e^{-4x}$

$\Rightarrow e^{2x} = -4$

This has no real solutions, so there are no stationary points.



4) d)

$y = (2x - 1)\mathrm{e}^{-2x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -2(2x - 1)\mathrm{e}^{-2x} + 2\mathrm{e}^{-2x}$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$-2(2x - 1)\mathrm{e}^{-2x} + 2\mathrm{e}^{-2x} = 0$

$\Rightarrow (-4x + 2)\mathrm{e}^{-2x} + 2\mathrm{e}^{-2x} = 0$

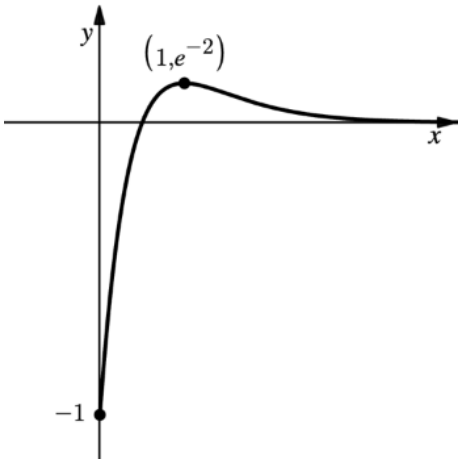
$\Rightarrow (-4x + 2 + 2)\mathrm{e}^{-2x} = 0$

$\Rightarrow (4 - 4x)\mathrm{e}^{-2x} = 0$

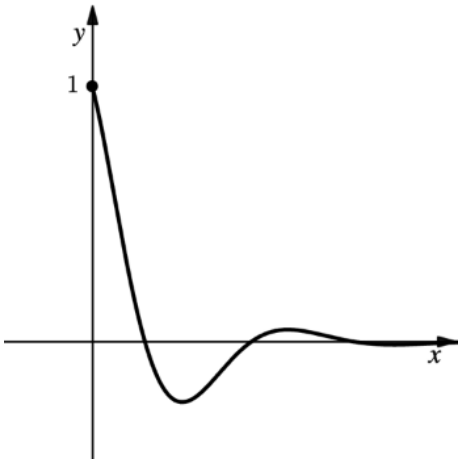
$\Rightarrow x = 1$

$y|_{x=1} = \mathrm{e}^{-2}$

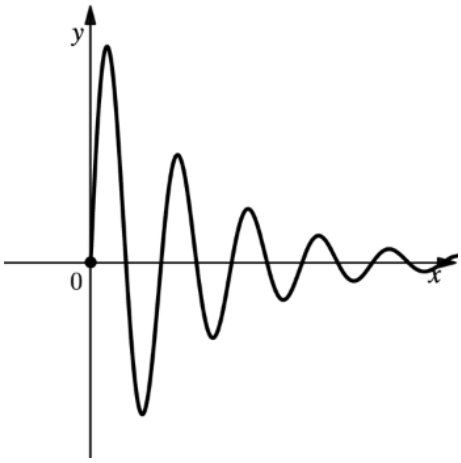
So there is a stationary point at $(1, \mathrm{e}^{-2})$



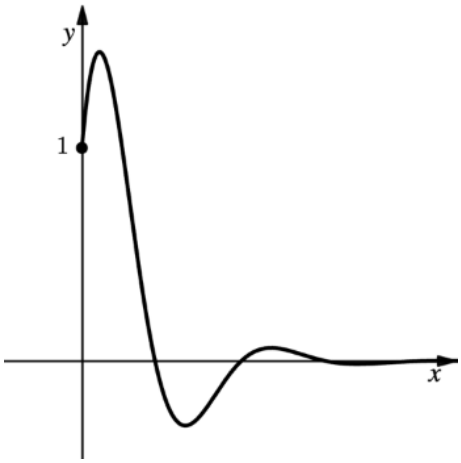
5) a)



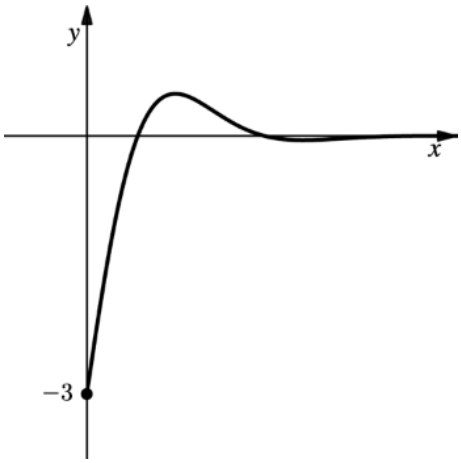
5) b)



5) c)



5) d)



6) a)

$2\ddot{x} + 5\dot{x} + 3x = 0$

Auxiliary Equation:

$2m^2 + 5m + 3 = 0$

$\Rightarrow (2m + 3)(m + 1) = 0$

$\Rightarrow m = -\frac{3}{2}$ or $m = -1$

General Solution:

$x = Ae^{-\frac{3}{2}t} + Be^{-t}$

Substitute $x = 2$ and $t = 0$:

$2 = Ae^0 + Be^0$

$\Rightarrow A + B = 2$ (1)

$\dot{x} = -\frac{3}{2}Ae^{-\frac{3}{2}t} - Be^{-t}$

Substitute $\dot{x} = 1$ and $t = 0$:

$1 = -\frac{3}{2}Ae^0 - Be^0$

$\Rightarrow -\frac{3}{2}A - B = 1$ (2)

Solve (1) & (2):

$A = -6$ and $B = 8$

Particular Solution:

$x = -6e^{-\frac{3}{2}t} + 8e^{-t}$

6) b)

This is Heavy Damping (Overdamping) as the auxiliary equation has two distinct real roots.

6) c)

$x = -6e^{-\frac{3}{2}t} + 8e^{-t}$

$\dot{x} = 9e^{-\frac{3}{2}t} - 8e^{-t}$

Stationary points exist when $\dot{x} = 0$:

$0 = 9e^{-\frac{3}{2}t} - 8e^{-t}$

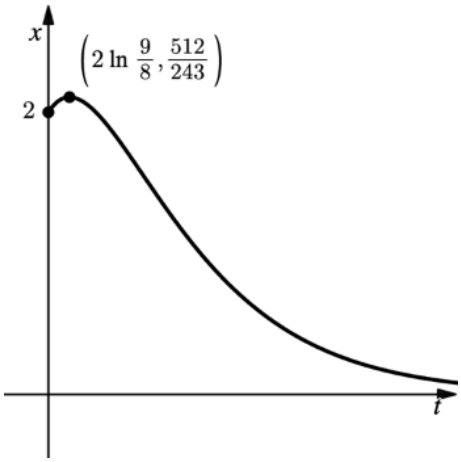
$\Rightarrow 8e^{-t} = 9e^{-\frac{3}{2}t}$

$\Rightarrow e^{\frac{1}{2}t} = \frac{9}{8}$

$\Rightarrow \frac{1}{2}t = \ln\frac{9}{8}$

$\Rightarrow t = 2\ln\frac{9}{8}$

6) d)



7) a)

$4\ddot{x} + 4\dot{x} + 5x = 0$

Auxiliary Equation:

$4m^2 + 4m + 5 = 0$

$\Rightarrow 4(m^2 + m) + 5 = 0$

$\Rightarrow 4\left(\left(m + \frac{1}{2}\right)^2 - \frac{1}{4}\right) + 5 = 0$

$\Rightarrow 4\left(m + \frac{1}{2}\right)^2 - 1 + 5 = 0$

$\Rightarrow \left(m + \frac{1}{2}\right)^2 = -1$

$\Rightarrow m = -\frac{1}{2} \pm \mathrm{i}$

General Solution:

$x = \mathrm{e}^{-\frac{1}{2}t}(A\cos t + B\sin t)$

Substitute $x = 2$ and $t = 0$:

$2 = \mathrm{e}^0(A\cos 0 + B\sin 0)$

$\Rightarrow A = 2$

$\therefore x = \mathrm{e}^{-\frac{1}{2}t}(2\cos t + B\sin t)$

$\dot{x} = \mathrm{e}^{-\frac{1}{2}t}(-2\sin t + B\cos t) - \frac{1}{2}\mathrm{e}^{-\frac{1}{2}t}(2\cos t + B\sin t)$

Substitute $\dot{x} = -1$ and $t = 0$:

$-1 = \mathrm{e}^0(-2\sin 0 + B\cos 0) - \frac{1}{2}\mathrm{e}^0(2\cos 0 + B\sin 0)$

$\Rightarrow -1 = (0 + B) - \frac{1}{2}(2 + 0)$

$\Rightarrow -1 = B - 1$

$\Rightarrow B = 0$

Particular Solution:

$x = 2\mathrm{e}^{-\frac{1}{2}t}\cos t$

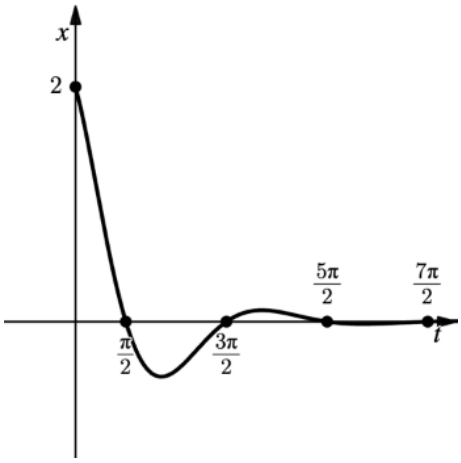
7) b)

This is Light Damping (Underdamping) as the auxiliary equation has complex conjugate solutions.

7) c)

$\cos t$ has period 2π, so the period of the oscillations is 2π

7) d)



8) a)

$y = 4e^{-3x} - 2e^{-2x} + 1$

$\frac{dy}{dx} = -12e^{-3x} + 4e^{-2x}$

Stationary points exist when $\frac{dy}{dx} = 0$

$-12e^{-3x} + 4e^{-2x} = 0$

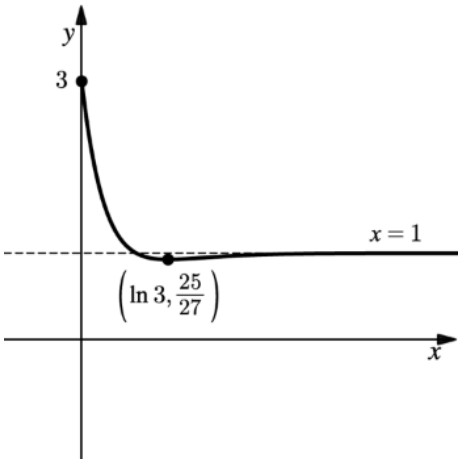
$\Rightarrow e^{-2x} = 3e^{-3x}$

$\Rightarrow e^{x} = 3$

$\Rightarrow x = \ln 3$

$y|_{x=\ln 3} = \frac{25}{27}$

So there is a stationary point at $\left(\ln 3, \frac{25}{27}\right)$



8) b)

$y = \mathrm{e}^{-2x} - 2\mathrm{e}^{-3x} - 2$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -2\mathrm{e}^{-2x} + 6\mathrm{e}^{-3x}$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$-2\mathrm{e}^{-2x} + 6\mathrm{e}^{-3x} = 0$

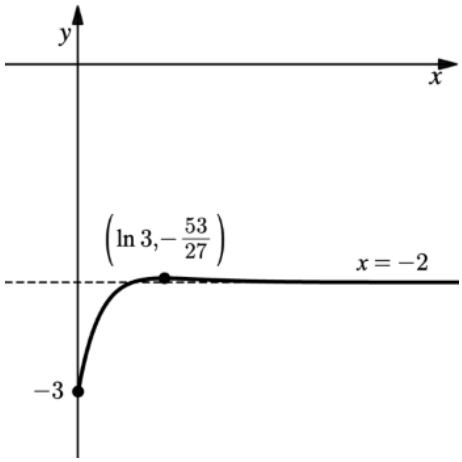
$\Rightarrow 3\mathrm{e}^{-3x} = \mathrm{e}^{-2x}$

$\Rightarrow 3 = \mathrm{e}^{x}$

$\Rightarrow x = \ln 3$

$y|_{x=\ln 3} = -\frac{53}{27}$

So there is a stationary point at $\left(\ln 3, -\frac{53}{27}\right)$



8) c)

$y = 3e^{-x} - 3e^{-2x} - 1$

$\frac{dy}{dx} = -3e^{-x} + 6e^{-2x}$

Stationary points exist when $\frac{dy}{dx} = 0$

$-3e^{-x} + 6e^{-2x} = 0$

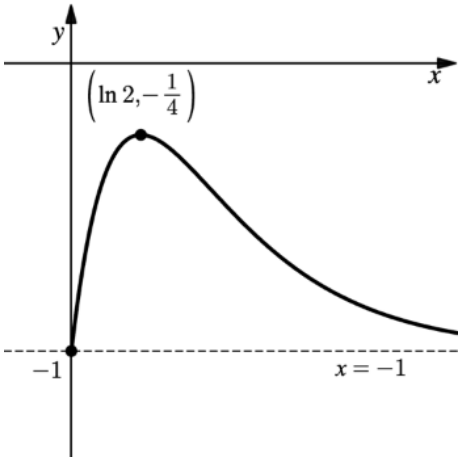
$\Rightarrow 2e^{-2x} = e^{-x}$

$\Rightarrow 2 = e^{x}$

$\Rightarrow x = \ln 2$

$y|_{x=\ln 2} = -\frac{1}{4}$

So there is a stationary point at $\left(\ln 2, -\frac{1}{4}\right)$



8) d)

$y = (3 - 9x)\mathrm{e}^{-3x} + 2$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -3(3 - 9x)\mathrm{e}^{-3x} - 9\mathrm{e}^{-3x}$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$-3(3 - 9x)\mathrm{e}^{-3x} - 9\mathrm{e}^{-3x} = 0$

$\Rightarrow (3 - 9x)\mathrm{e}^{-3x} + 3\mathrm{e}^{-3x} = 0$

$\Rightarrow (3 - 9x + 3)\mathrm{e}^{-3x} = 0$

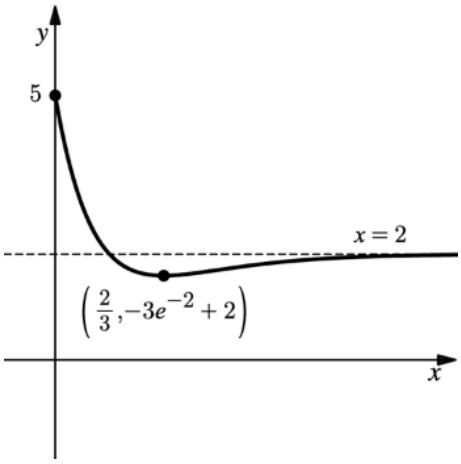
$\Rightarrow (6 - 9x)\mathrm{e}^{-3x} = 0$

$\Rightarrow 6 - 9x = 0$

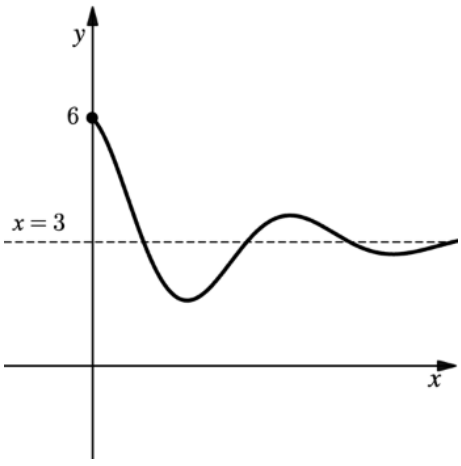
$\Rightarrow x = \frac{2}{3}$

$y|_{x=\frac{2}{3}} = -3\mathrm{e}^{-2} + 2$

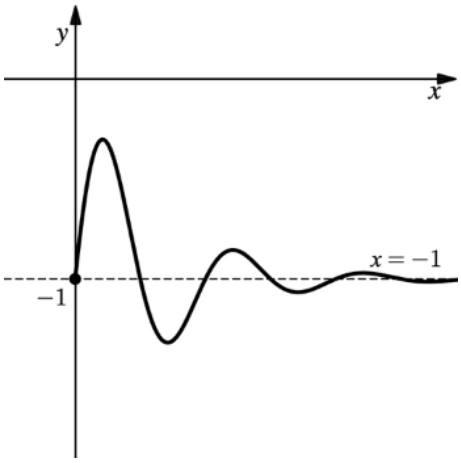
So there is a stationary point at $\left(\frac{2}{3}, -3\mathrm{e}^{-2} + 2\right)$



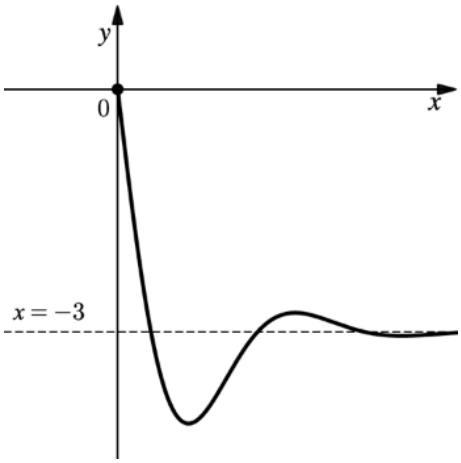
9) a)



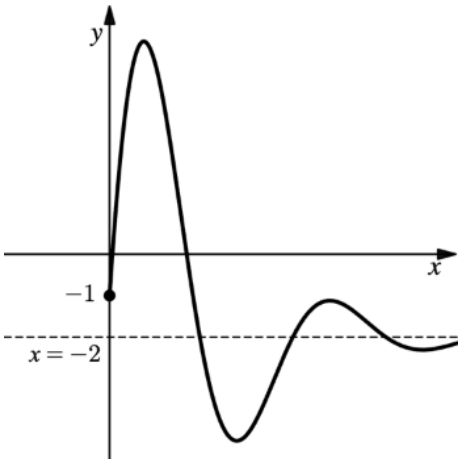
9) b)



9) c)



9) d)



10) a)

$\ddot{x} + 6\dot{x} + 8x = 8$

Homogeneous Equation:

$\ddot{x} + 6\dot{x} + 8x = 0$

Auxiliary Equation:

$m^2 + 6m + 8 = 0$

$\Rightarrow (m + 2)(m + 4) = 0$

$\Rightarrow m = -2$ or $m = -4$

Complementary Function:

$x = Ae^{-2t} + Be^{-4t}$

Try this Particular Integral:

$x = k$

$\dot{x} = 0$

$\ddot{x} = 0$

$\ddot{x} + 6\dot{x} + 8x = 8$

$\Rightarrow (0) + 6(0) + 8(k) = 8$

$\Rightarrow k = 1$

Particular Integral:

$x = 1$

General Solution:

$x = Ae^{-2t} + Be^{-4t} + 1$

Substitute $x = 2$ and $t = 0$:

$2 = Ae^0 + Be^0 + 1$

$\Rightarrow A + B = 1$ (1)

$\dot{x} = -2Ae^{-2t} - 4Be^{-4t}$

Substitute $\dot{x} = 2$ and $t = 0$:

$2 = -2Ae^0 - 4Be^0$

$\Rightarrow -2A - 4B = 2$ (2)

Solve (1) & (2):

$A = 3$ and $B = -2$

Particular Solution:

$x = 3e^{-2t} - 2e^{-4t} + 1$

10) b)

This is Heavy Damping (Overdamping) as the auxiliary equation has two distinct real roots.

10) c)

$x = 3e^{-2t} - 2e^{-4t} + 1$

$\dot{x} = -6e^{-2t} + 8e^{-4t}$

Stationary points exist when $\dot{x} = 0$:

$0 = -6e^{-2t} + 8e^{-4t}$

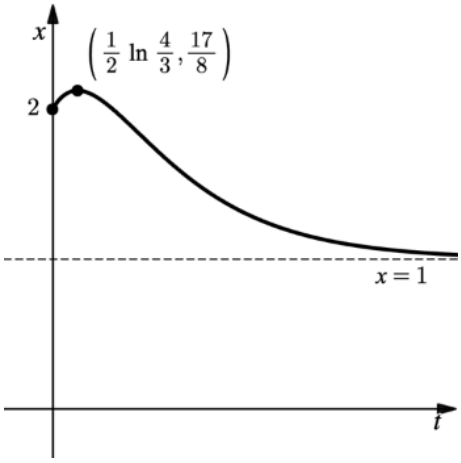
$\Rightarrow 3e^{-2t} = 4e^{-4t}$

$\Rightarrow e^{2t} = \frac{4}{3}$

$\Rightarrow 2t = \ln\frac{4}{3}$

$\Rightarrow t = \frac{1}{2}\ln\frac{4}{3}$

10) d)



11) a)

$16\ddot{x} + 8\dot{x} + x = -3$

Homogeneous Equation:

$16\ddot{x} + 8\dot{x} + x = 0$

Auxiliary Equation:

$16m^2 + 8m + 1 = 0$

$\Rightarrow (4m + 1)^2 = 0$

$\Rightarrow m = -\frac{1}{4}$

Complementary Function:

$x = Ate^{-\frac{1}{4}t} + Be^{-\frac{1}{4}t} = (At + B)e^{-\frac{1}{4}t}$

Try this Particular Integral:

$x = k$

$\dot{x} = 0$

$\ddot{x} = 0$

$16\ddot{x} + 8\dot{x} + x = -3$

$\Rightarrow 16(0) + 8(0) + (k) = -3$

$\Rightarrow k = -3$

Particular Integral:

$x = -3$

General Solution:

$x = (At + B)e^{-\frac{1}{4}t} - 3$

Substitute $x = 1$ and $t = 0$:

$1 = (A \times 0 + B)e^0 - 3$

$\Rightarrow 1 = B - 3$

$\Rightarrow B = 4$

$\therefore x = (At + 4)e^{-\frac{1}{4}t} - 3$

$\dot{x} = -\frac{1}{4}(At + 4)e^{-\frac{1}{4}t} + Ae^{-\frac{1}{4}t}$

Substitute $\dot{x} = 1$ and $t = 0$:

$1 = -\frac{1}{4}(A \times 0 + 4)e^0 + Ae^0$

$\Rightarrow 1 = -1 + A$

$\Rightarrow A = 2$

Particular Solution:

$x = (2t + 4)e^{-\frac{1}{4}t} - 3$

11) b)

This is Critical Damping as the auxiliary equation has two repeated real roots.

11) c)

$x = (2t + 4)\mathrm{e}^{-\frac{1}{4}t} - 3$

$\dot{x} = -\frac{1}{4}(2t + 4)\mathrm{e}^{-\frac{1}{4}t} + 2\mathrm{e}^{-\frac{1}{4}t}$

Stationary points exist when $\dot{x} = 0$:

$0 = -\frac{1}{4}(2t + 4)\mathrm{e}^{-\frac{1}{4}t} + 2\mathrm{e}^{-\frac{1}{4}t}$

$\Rightarrow 0 = -\frac{1}{4}(t + 2)\mathrm{e}^{-\frac{1}{4}t} + \mathrm{e}^{-\frac{1}{4}t}$

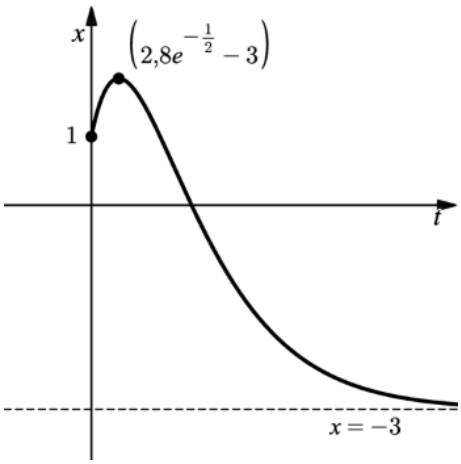
$\Rightarrow 0 = (t + 2)\mathrm{e}^{-\frac{1}{4}t} - 4\mathrm{e}^{-\frac{1}{4}t}$

$\Rightarrow 0 = (t + 2 - 4)\mathrm{e}^{-\frac{1}{4}t}$

$\Rightarrow 0 = (t - 2)\mathrm{e}^{-\frac{1}{4}t}$

$\Rightarrow t = 2$

11) d)



12) a)

$4\ddot{x} + 8\dot{x} + 13x = 26$

Homogeneous Equation:

$4\ddot{x} + 8\dot{x} + 13x = 0$

Auxiliary Equation:

$4m^2 + 8m + 13 = 0$

$\Rightarrow 4(m^2 + 2m) + 13 = 0$

$\Rightarrow 4((m+1)^2 - 1) + 13 = 0$

$\Rightarrow 4(m+1)^2 - 4 + 13 = 0$

$\Rightarrow (m+1)^2 = -\frac{9}{4}$

$\Rightarrow m = -1 \pm \frac{3}{2}\mathrm{i}$

Complementary Function:

$x = \mathrm{e}^{-t}\left(A\cos\frac{3}{2}t + B\sin\frac{3}{2}t\right)$

Try this Particular Integral:

$x = k$

$\dot{x} = 0$

$\ddot{x} = 0$

$4\ddot{x} + 8\dot{x} + 13x = 26$

$\Rightarrow 4(0) + 8(0) + 13(k) = 26$

$\Rightarrow k = 2$

Particular Integral:

$x = 2$

General Solution:

$x = \mathrm{e}^{-t}\left(A\cos\frac{3}{2}t + B\sin\frac{3}{2}t\right) + 2$

Substitute $x = 1$ and $t = 0$:

$1 = \mathrm{e}^0(A\cos 0 + B\sin 0) + 2$

$\Rightarrow 1 = A + 2$

$\Rightarrow A = -1$

$\therefore x = \mathrm{e}^{-t}\left(-\cos\frac{3}{2}t + B\sin\frac{3}{2}t\right) + 2$

$\dot{x} = \mathrm{e}^{-t}\left(\frac{3}{2}\sin\frac{3}{2}t + \frac{3}{2}B\cos\frac{3}{2}t\right) - \mathrm{e}^{-t}\left(-\cos\frac{3}{2}t + B\sin\frac{3}{2}t\right)$

Substitute $\dot{x} = 4$ and $t = 0$:

$4 = \mathrm{e}^0\left(\frac{3}{2}\sin 0 + \frac{3}{2}B\cos 0\right) - \mathrm{e}^0(-\cos 0 + B\sin 0)$

$\Rightarrow 4 = \left(0 + \frac{3}{2}B\right) - (-1 + 0)$

$\Rightarrow 4 = \frac{3}{2}B + 1$

$\Rightarrow B = 2$

Particular Solution:

$x = \mathrm{e}^{-t}\left(-\cos\frac{3}{2}t + 2\sin\frac{3}{2}t\right) + 2$

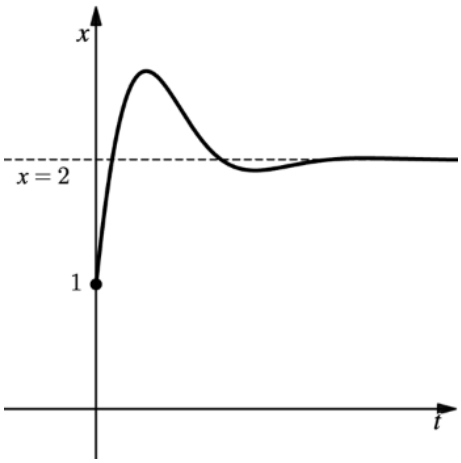
12) b)

This is Light Damping (Underdamping) as the auxiliary equation has complex conjugate solutions.

12) c)

$\cos\frac{3}{2}t$ and $\sin\frac{3}{2}t$ have period $\frac{2\pi}{\frac{3}{2}}$, so the period of the oscillations is $\frac{4\pi}{3}$

12) d)



13) a)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 8\frac{\mathrm{d}y}{\mathrm{d}x} + 17y = 40\cos x$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 8\frac{\mathrm{d}y}{\mathrm{d}x} + 17y = 0$

Auxiliary Equation:

$m^2 + 8m + 17 = 0$

$\Rightarrow (m+4)^2 - 16 + 17 = 0$

$\Rightarrow (m+4)^2 = -1$

$\Rightarrow m = -4 \pm \mathrm{i}$

Complementary Function:

$y = \mathrm{e}^{-4x}(A\cos x + B\sin x)$

Try this Particular Integral:

$y = p\cos x + q\sin x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -p\sin x + q\cos x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -p\cos x - q\sin x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 8\frac{\mathrm{d}y}{\mathrm{d}x} + 17y = 40\cos x$

$\Rightarrow (-p\cos x - q\sin x) + 8(-p\sin x + q\cos x) + 17(p\cos x + q\sin x) = 40\cos x$

$\Rightarrow (-p + 8q + 17p)\cos x + (-q - 8p + 17q)\sin x = 40\cos x$

$\Rightarrow (16p + 8q)\cos x + (-8p + 16q)\sin x = 40\cos x$

Compare coefficients:

$16p + 8q = 40$ (1)

$-8p + 16q = 0$ (2)

Solve (1) & (2):

$p = 2$ and $q = 1$

Particular Integral:

$y = 2\cos x + \sin x$

General Solution:

$y = \mathrm{e}^{-4x}(A\cos x + B\sin x) + 2\cos x + \sin x$

Substitute $y = 2$ and $x = 0$:

$2 = \mathrm{e}^0(A\cos 0 + B\sin 0) + 2\cos 0 + \sin 0$

$\Rightarrow 2 = A + 2 + 0 \Rightarrow A = 0$

$\therefore y = B\mathrm{e}^{-4x}\sin x + 2\cos x + \sin x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = B\mathrm{e}^{-4x}\cos x - 4B\mathrm{e}^{-4x}\sin x - 2\sin x + \cos x$

Substitute $\frac{\mathrm{d}y}{\mathrm{d}x} = 2$ and $x = 0$:

$2 = B\mathrm{e}^0 \cos 0 - 4B\mathrm{e}^0 \sin 0 - 2\sin 0 + \cos 0$

$\Rightarrow 2 = B - 0 - 0 + 1 \Rightarrow B = 1$

Particular Solution:

$y = \mathrm{e}^{-4x} \sin x + 2\cos x + \sin x$

13) b)

As $x \to \infty$, $\mathrm{e}^{-4x} \to 0$, so $y \to 2\cos x + \sin x$, which is simple harmonic form.

14) a)

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 9y = \frac{9}{2}x$

Homogeneous Equation:

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 9y = 0$

Auxiliary Equation:

$4m^2 + 9 = 0$

$\Rightarrow m^2 = -\frac{9}{4}$

$\Rightarrow m = \pm\frac{3}{2}\mathrm{i}$

Complementary Function:

$y = A\cos\frac{3}{2}x + B\sin\frac{3}{2}x$

Try this Particular Integral:

$y = px + q$

$\frac{\mathrm{d}y}{\mathrm{d}x} = p$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 9y = \frac{9}{2}x$

$\Rightarrow 4(0) + 9(px + q) = \frac{9}{2}x$

$\Rightarrow (9p)x + (9q) = \frac{9}{2}x$

Compare coefficients:

$9p = \frac{9}{2} \Rightarrow p = \frac{1}{2}$

$9q = 0 \Rightarrow q = 0$

Particular Integral:

$y = \frac{1}{2}x$

General Solution:

$y = A\cos\frac{3}{2}x + B\sin\frac{3}{2}x + \frac{1}{2}x$

Substitute $y = 4$ and $x = 0$:

$4 = A\cos 0 + B\sin 0 + \frac{1}{2}(0)$

$\Rightarrow 4 = A + 0 + 0 \Rightarrow A = 4$

$\therefore y = 4\cos\frac{3}{2}x + B\sin\frac{3}{2}x + \frac{1}{2}x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -6\sin\frac{3}{2}x + \frac{3}{2}B\cos\frac{3}{2}x + \frac{1}{2}$

Substitute $\frac{\mathrm{d}y}{\mathrm{d}x} = -1$ and $x = 0$:

$-1 = -6\sin 0 + \frac{3}{2}B\cos 0 + \frac{1}{2}$

$\Rightarrow -1 = 0 + \frac{3}{2}B + \frac{1}{2} \Rightarrow \frac{3}{2}B = -\frac{3}{2} \Rightarrow B = -1$

Particular Solution:

$y = 4\cos\frac{3}{2}x - \sin\frac{3}{2}x + \frac{1}{2}x$

14) b)

$\frac{\mathrm{d}y}{\mathrm{d}x} = -6\sin\frac{3}{2}x - \frac{3}{2}\cos\frac{3}{2}x + \frac{1}{2}$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\Rightarrow -6\sin\frac{3}{2}x - \frac{3}{2}\cos\frac{3}{2}x + \frac{1}{2} = 0$

$\Rightarrow 12\sin\frac{3}{2}x + 3\cos\frac{3}{2}x - 1 = 0$

$\Rightarrow 3\sqrt{17}\sin\left(\frac{3}{2}x + \arctan\frac{1}{4}\right) = 1$

$\Rightarrow \sin\left(\frac{3}{2}x + \arctan\frac{1}{4}\right) = \frac{1}{3\sqrt{17}}$

$\Rightarrow \frac{3}{2}x + \arctan\frac{1}{4} = \arcsin\left(\frac{1}{3\sqrt{17}}\right)$

$\Rightarrow x = \frac{2}{3}\left(\arcsin\left(\frac{1}{3\sqrt{17}}\right) - \arctan\frac{1}{4}\right)$

$= -0.109$ (3 SF)

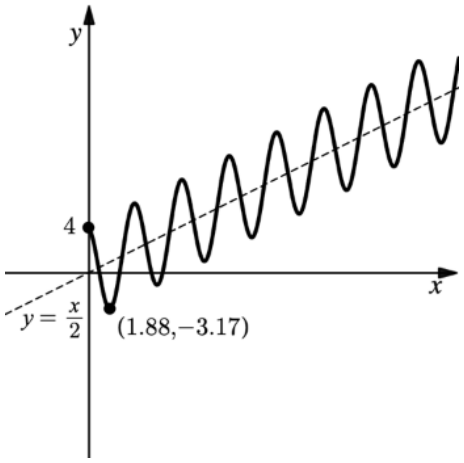
We need $x \geq 0$ so:

$\Rightarrow \frac{3}{2}x + \arctan\frac{1}{4} = \pi - \arcsin\left(\frac{1}{3\sqrt{17}}\right)$

$\Rightarrow x = \frac{2}{3}\left(\pi - \arcsin\left(\frac{1}{3\sqrt{17}}\right) - \arctan\frac{1}{4}\right)$

$= 1.88$ (3 SF)

14) c)



14) d)

As the amplitude is decreasing, the auxiliary equation must have complex roots with negative real part. $4\cos\frac{3}{2}x - \sin\frac{3}{2}x$ has a fixed amplitude and multiplying this by e^{kx} where $k < 0$ ensures that the amplitude of the curve decreases as $x \to \infty$

14) e)

$y = e^{kx}\left(4\cos\frac{3}{2}x - \sin\frac{3}{2}x\right) + \frac{1}{2}x$

$\Rightarrow \frac{dy}{dx} = e^{kx}\left(-6\sin\frac{3}{2}x - \frac{3}{2}\cos\frac{3}{2}x\right) + ke^{kx}\left(4\cos\frac{3}{2}x - \sin\frac{3}{2}x\right) + \frac{1}{2}$

$\left.\frac{dy}{dx}\right|_{x=0} = e^{0}\left(-6\sin 0 - \frac{3}{2}\cos 0\right) + ke^{0}(4\cos 0 - \sin 0) + \frac{1}{2}$

$= \left(0 - \frac{3}{2}\right) + k(4 - 0) + \frac{1}{2}$

$= -\frac{3}{2} + 4k + \frac{1}{2}$

$= 4k - 1$

If $4k - 1 = -1$ then $k = 0$

As $k < 0$, we cannot have that $\left.\frac{dy}{dx}\right|_{x=0} = -1$

14) f)

